# Bradykinesia Analysis Workflow

**Data Organization**

All data must be stored in one folder. All files in the folder must be titled according to the folder name ("FolderName") with extensions describing the contents of each file. The contents of the folder are:

- Original video of the UPDRS examination titled "FolderName"
- MediaPipe output video titled "FolderName_Pose_MediaPipe.avi"
- MediaPipe body keypoint outputs tilted "FolderName_Body_Outputs.csv"
- MediaPIpe hand keypoint outputs titled "FolderName_Hand_Outputs.csv"
- OpenPose output video titled "FolderName_Pose_OpenPose.avi"
- Sub-folder titled "JSON_Files" containing each OpenPose output JSON file
- Spreadsheet titled "FolderName_Segmentation.xlsx" containing start and end frames of each bradykinesia assessment; template spreadsheet is included in folder titled "FolderName"

**Workflow**

1. Run *PD_BradyKinesia_Analysis.m* in MATLAB; this function calls the subsequent functions in succession.

2. Sub-function *load_pose.m* prompts the user to select a folder. Videos and associated pose estimation data from the folder are loaded. Pose estimation data are segmented based on start and end frames from the Excel spreadsheet in the selected folder (specified in line 391).

3. Sub-function *gapFill_filter.m* automatically gap fills missing pose data using linear interpolation for gaps up to 0.12 s and applies a low-pass filter using a 4$^{th}$ order Butterworth filter with a cut-off frequency of 5 Hz.

4. *findEvents.m* opens a graphical user interface that lets the user inspect and correct automatically detected events used to identify movement cycles of the bradykinesia tasks (Fig. 1). There are subplots with time-series corresponding to left and right limbs for each bradykinesia task. The y-axis for subplots of:
   - finger tapping, show the Euclidean distance between the low-pass filtered, MediaPipe-tracked keypoints of the tips of the index and thumb
   - hand open/close, show the Euclidean distance between the low-pass filtered, MediaPipe-tracked keypoints of the tips of the index and thumb
   - hand pronation/supination, show the vertical coordinate of the low-pass filtered, MediaPipe-tracked keypoint of the tip of the thumb
   - toe tapping, show the vertical coordinate of the OpenPose-tracked big toe keypoint
   - leg agility, show the vertical coordinate of the OpenPose-tracked heel keypoint

Movement cycle events are automatically determined based on:

- finger tapping, the frames in which the keypoints of tips of the index and thumb are furthest apart
- hand open/close, the frames in which the keypoints of tips of the index and thumb are furthest apart
- hand pronation/supination, the frames in which the keypoint of the tip of the thumb is at its highest vertical position
- toe tapping, the frames in which the keypoint of big toe is at its highest vertical position
- leg agility, the frames in which the keypoint of the heel is at its highest vertical position

Use slider or use cursor to update the image to show the selected frame; selected frame number is indicated in the title of image and a grey background bar in the relevant subplot to which the frame belongs. The image can be toggled to show either MediaPipe or OpenPose tracking using radio buttons located in the lower left side. To delete spurious events: use brush tool to select event(s) and click 'Delete Events'. To create missing events: use cursor to select frame and click 'Create Events' to create new event. For a quick visual check of whether events have been selected correctly, click 'Summary Plot' which opens a new window with overlaid curves for each movement cycle (Fig. 2); the curves are expected to follow a repeatable pattern (with some degree of variability) when events have been selected correctly. Click 'Save' when done.

5. *calcMovementParameters.m* automatically calculates temporal and spatial movement parameters. Temporal parameters are based on the time between event times. The mean movement time and the standard deviation of movement time reflect the average and the variability of movement time. Spatial parameters are based on the amplitude of the movements, described below for each bradykinesia item:
   - finger tapping, the pixel distance between the tips of the index and thumb at the event times normalized to torso height (the pixel distance between the Neck and midhip OpenPose keypoints).
   - hand open/close, the mean pixel distance between the tips of the index, middle, ring and little finger relative to the tip of the thumb at event times normalized to torso height.
   - hand pronation/supination, no amplitude value is calculated because the movement is expressed predominantly in a circular motion.
   - toe tapping, the vertical pixel distance between the keypoint of the big toe of the active (tapping) foot relative to the keypoint of the big toe of the inactive (stationary) foot at event times normalized to torso height.
   - leg agility, the vertical pixel distance between the keypoint of the heel of the active (moving) foot relative to the keypoint of the heel of the inactive (stationary) foot at event times normalized to torso height.

The mean, standard deviation and slope (based on a linear model fit) of amplitude are calculated. These spatial parameters reflect the average and the variability of amplitude, and changes in amplitude over the duration of the bradykinesia tests.

6. Contents saved in MATLAB data file
The output 'FolderName.mat' file is saved in the selected folder and contains 10 structures that are briefly described below:
   - amplitudeParameters
     Struct with amplitude variables for bradykinesia tasks (excluding hand pronation/supination).
   - auto_events
     Struct with frame numbers of events for all bradykinesia tasks. Note that frame numbers refer to each segments (i.e., a frame number of 10 refers to the tenth frame within a specific segment).
   - bradyKin_analyze
     Struct with logical variables. Variables are true by default. Variables can manually be set to false if only a portion of the bradykinesia tasks are performed.
   - bradyKin_segments
     Struct with start and end frames of all bradykinesia tasks. Frame numbers are imported from the Excel spreadsheet containing start and end frames.
   - data
     Struct containing pose data for OpenPose or MediaPipe.
   - frameInfo
     Struct containing logical vectors indicating whether the pose data have been gap filled.
   - output_name
     Name of selected folder.
   - temporalParameters
     Struct with temporal variables for bradykinesia tasks.
   - timeStamp
     Vector containing time stamps for each frame in original video.
   - videoInfo
     Struct containing VideoReader variables for original video and output videos for OpenPose and MediaPipe. The VideoReader variables contain meta-information of videos (e.g., frame rate and pixel resolution).

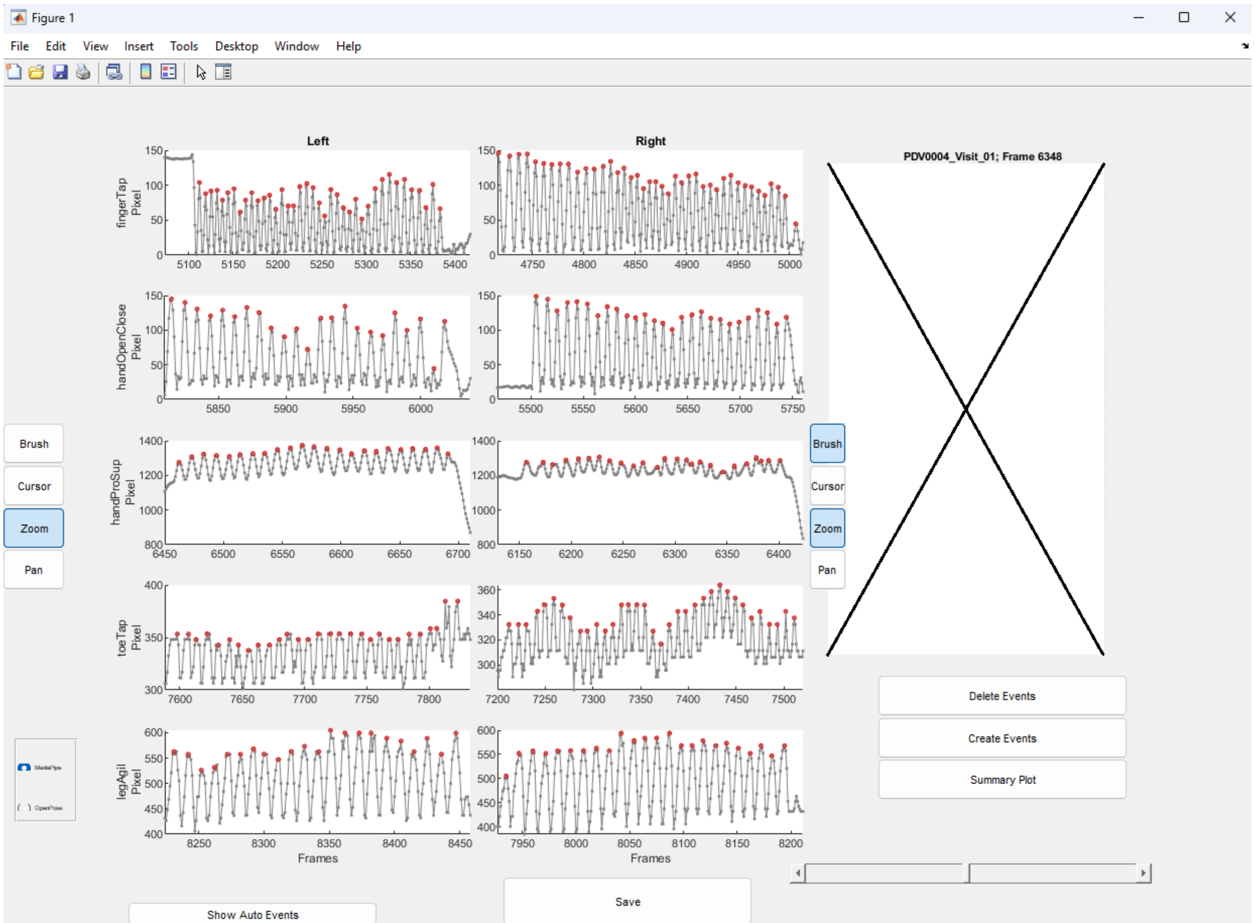


Figure 1. Graphical user interface to visually inspect and correct events of movement cycles.

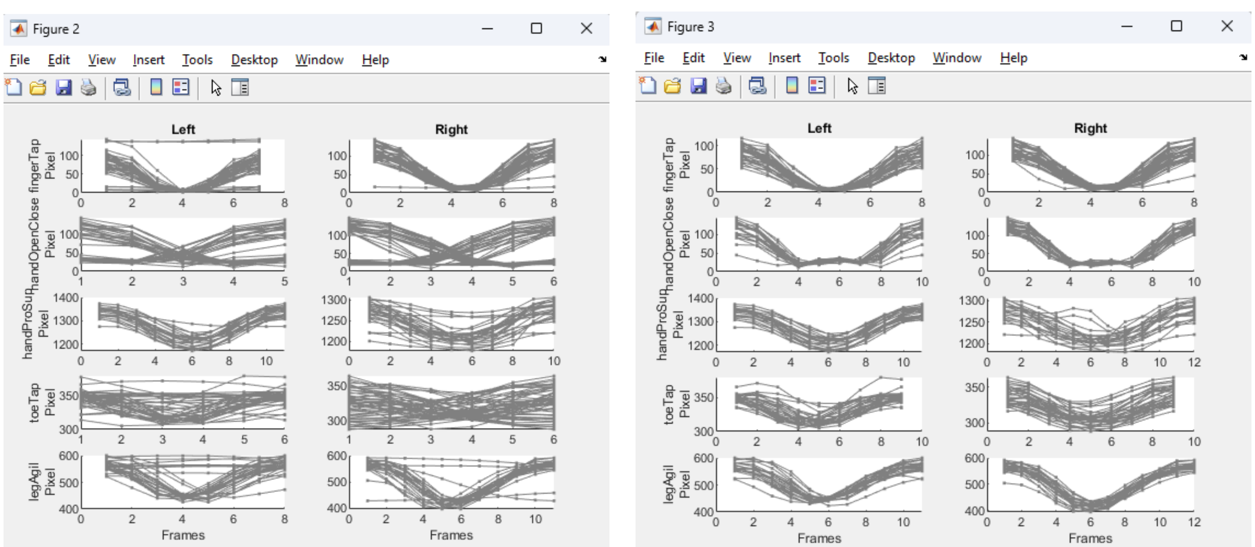


Figure 2. Summary plots to confirm whether events have been correctly selected. Left plot shows overlaid curves for each movement cycle *before* events have been correct; right plot shows curves *after* events have been corrected. Note that curves on right plot are similar (with some degree of variability) while curves on left show considerably more variability.